Weichen Huang

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Research Statement

In recent years, LLM-based text-to-fMRI encoding models have approached the noise ceiling on several neural predictivity benchmarks. My research investigates **why hidden activation structure in LLMs is so predictive of human brain activity** and what that implies for better understanding both **LLM representations** and the **human language network**. I routinely use tools from *mechanistic interpretability* (e.g., sparse autoencoders, linear/nonlinear probes) and *computational neuroscience* (e.g., representational similarity analysis, representational geometry) to test mechanistic hypotheses about which **features and transformations** drive brain alignment.

Research Experience

- **Language Intelligence Lab @ Georgia Tech** — Atlanta, GA
  Language Network Research / with PI: Dr. Anna Ivanova — *Oct 2025 – Present*
  - Investigating how to interpret large language model based fMRI encoding models using mechanistic interpretability approaches like TunedLens, prompt recovery from hidden activations and computational neuroscience approaches like Representational Similarity Analysis.
  - The primary research goal is to build more fine grained semantic maps of the language network of the human brain.
- **DiCarlo Lab @ MIT (during RSI 2024)** — *(remote/visiting)* Cambridge, MA
  Neuromorphic computational vision / with Yudi Xie and PI: Dr. James J. DiCarlo — *Jun 2024 – Oct 2024*
  - Continued neuromorphic computing research following RSI summer school on convergent neural representations.
  - Investigated the role of the primate ventral visual stream in estimating spatial latents, challenging a purely object-categorization framing.
  - Paper published to ICLR 2025: Vision Models Trained to Estimate Spatial Latents Learn Ventral-Stream-Aligned Representations.
- **ML in Medical Imaging Lab @ UCD** — Dublin, Ireland
  Deep learning for medical imaging and multimodal diagnosis / with PI: Dr. Kathleen Curran — *Feb 2023 – Jan 2025*
  - Studied spurious correlations in chest X-ray (CXR) interpretation models; applied reinforcement learning and self-/semi-supervised training to improve robustness.
  - Developed MRI–PET fusion technique with dynamic image compression for Alzheimer's disease diagnosis; evaluated on ADNI with improvements over prior baselines.
  - Built multimodal contrastive learning framework combining MRI images with tabular clinical features via tabular attention; improved Alzheimer's prediction accuracy to 83.8%+.
  - Proposed graph-based contrastive learning framework enabling learning from an arbitrary number of modalities, improving generalizability.
  - Research published in IMVIP 2023, ICCV 2023; NeurIPS 2024.

Publications & Preprints

- Xie, Y., **Huang, W.**, Alter, E., Schwartz, J., Tenenbaum, J., DiCarlo, J. (2025). *Vision Models Trained to Estimate Spatial Latents Learned Similar Ventral-stream-aligned Representations.* Submitted to ICLR 2025 (under review).
- **Huang, W.** (2024). *Multimodal Representation Learning using Adaptive Graph Construction.* NeurIPS 2024 (High School Projects track; top 4/335). arXiv:2410.06395.
- **Huang, W.** (2023). *Multimodal Contrastive Learning and Tabular Attention for Automated Alzheimer's Disease Prediction.* ICCV 2023, pp. 2465–2474. arXiv:2308.15469.
- **Huang, W.**, Curran, K. (2023). *Spurious Correlation Mitigation in CXR Images via Reinforcement Learning and Self-Supervision.* IMVIP 2023. Zenodo.
- Chang, Z., **Huang, W.**, Curran, K. (2023). *Improved Multimodal Diagnosis of Alzheimer's Disease through MRI and PET Image Fusion.* RSNA 2023.
- **Huang, W.** (2022). *Feature Scaling and Attention Convolutions for Extreme Precipitation Prediction.* Fragile Earth 2022 (ACM SIGKDD workshop). PDF.

Education

- **Georgia Institute of Technology** — Atlanta, GA
  B.S. in Computer Science (Freshman, Class of 2029); GPA: 4.0/4.0 — *2025 – Present*
  - Selected coursework: Linear Algebra (MATH 1554), Applied Combinatorics (MATH 3012), Honors Multivariable Calculus (MATH 2561), General Psychology (PSYC 1101).
  - Competitive programming: Georgia Tech ICPC Regional (2025).
- **St Andrew's College** — Dublin, Ireland
  International Baccalaureate; 43/45; SAT 1580/1600; DELF (French) B1 — *Graduated Jun 2025*
  - HL: Mathematics Analysis & Approaches (7), Physics (7), Chemistry (7).

Additional Training

- **MIT OpenCourseWare** — Online
  Selected courses — *2022 – 2024*
  - 6.006 Introduction to Algorithms; 6.042J Mathematics for Computer Science; 6.046J Design and Analysis of Algorithms.
  - 18.01 Single Variable Calculus; 18.02 Multivariable Calculus; 18.03 Differential Equations; 18.657 Mathematics of Machine Learning.
- **Stanford University (Online)** — Online
  Mathematical Thinking; Logical notation and problem solving — *2021 – 2022*
  - Completed coursework focused on proof techniques and formal reasoning.

Leadership, Service & Reviewing

- **President, St Andrew's College Competitive Programming Club** (2022–Present): taught C++/Python, data structures and algorithms; organized participation in Bebras Computing Challenge and other contests.
- **Volunteering:** Dún Laoghaire Tidy Towns Group; International Conference on Computer Vision (Paris, 2023).
- **Conference Reviewer:** ICLR 2025.

Awards & Honors

- **Competitive Programming:** USACO Gold (since 2023); WEOI Irish National Team (2024); AIPO/IrlCPC finalist (2022–2024); Meta Hacker Cup Round 2 (2022); Google Kick Start Rounds E/F/G (2022); IOI Irish Team (2025).
- **Science Fairs:** Irish Research Council Award (BT Young Scientist Exhibition, 2024; Category 2nd place); SciFest Boston Scientific Award (2024); Stripe Software Award (BT Young Scientist Exhibition, 2023; Category 1st place); Met Éireann Award (BT Young Scientist Exhibition, 2022; Category 1st place); BT Young Scientist Exhibition Category 1st place (2021).
- **Other Academic:** BPhO Bronze (2024); Canadian Senior Math Contest Distinction and Roll of Honour (2024); CSMC Roll of Honour (2023); IrMO finalist (2022); St Andrew's College Academic Award (2021); TY Distinction (2023); SciFest Innovator's Award (2021).

Selected Projects & Demos

- **CXR Predict (demo).** Self-supervised cross-modal representation learning. Hugging Face Space.
- **Extreme weather prediction using attention-augmented convolutions.** Project page: weichen-huang.github.io/projects/saconvnets.
- **EfficientNet COVID-19 detection from lung X-ray imaging.** Project page: weichen-huang.github.io/projects/covidEfficientnet.

Skills

- **Programming-related:** NumPy, SciPy, PyTorch, C++, Java
- **Areas of Experience:** Mechanistic Interpretability; Representation Learning; Computer Vision and NLP; Computational Neuroscience; Medical Imaging; Competitive Programming